

Rikesh Mandal

Software Engineer | AI & Machine Learning | Backend Systems

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PROFESSIONAL SUMMARY

Software Engineer with an MSc in Artificial Intelligence (Distinction, 82%) and a BSc in Computer Networking & IT Security (First Class, 81.94%). Experienced in building and maintaining high-performance REST APIs and distributed backend systems in a regulated fintech environment. Deep practical expertise in machine learning, computer vision, and deep learning — including a Master's dissertation delivering a monocular image-to-3D reconstruction pipeline using PyTorch and neural network architectures. Strong Python and Java developer actively targeting software engineering roles at high-growth and world-class technology companies.

EXPERIENCE

Assistant Java Developer | *FocusOne Payment Solutions — Fintech*

2022 – 2024

Contributed to backend systems powering real-time credit and debit card authorisation for a regulated UK fintech platform.

- Designed and maintained high-performance REST APIs using Java and Spring Boot, contributing to a platform processing real-time card authorisations across thousands of daily transactions.
- Evaluated NATS messaging for event-driven architecture adoption, presenting findings that informed the team's infrastructure roadmap for low-latency, fault-tolerant distributed systems.
- Refactored legacy in-house applications, improving system reliability and reducing production incidents through targeted code improvements and root-cause debugging.
- Built and maintained comprehensive Swagger API documentation, cutting developer onboarding friction and improving cross-team collaboration on API contracts.
- Implemented new backend services aligned with evolving product requirements in a fast-paced Agile environment, consistently delivering features ahead of sprint deadlines.
- Authored system specifications and API design documentation, supporting long-term maintainability and regulatory compliance.

EDUCATION

MSc Artificial Intelligence — Distinction (82%) | *University of East London*

2024 – 2025

- Dissertation: Monocular Image-to-3D Reconstruction — end-to-end deep learning pipeline using a ResNet50 encoder-decoder for depth estimation, followed by 3D point cloud generation and Poisson surface reconstruction via Open3D.
- Modules included: Machine Learning, Deep Learning, Computer Vision, Neural Networks, Data Science, Research Methods.

BSc Computer Networking & IT Security — First Class (81.94%) | *London Metropolitan University*

2021 – 2023

- Covered: Network Architecture, Switching & Routing, Enterprise Security, Wireless Essentials, Network Automation.

KEY PROJECTS

Monocular Image-to-3D Reconstruction — [GitHub](#)

MSc dissertation project. Built a full deep learning pipeline: ResNet50 encoder-decoder trained on NYU Depth V2 dataset for monocular depth estimation from RGB images; converted depth maps to 3D point clouds using

Open3D; applied Poisson surface reconstruction and mesh refinement. Demonstrates end-to-end ML system design from data to deployable output.

Tech: *Python, PyTorch, OpenCV, Open3D, NumPy, Matplotlib*

Sign Language-to-Speech Glove — [GitHub](#)

Designed and built a hardware-software IoT system: flex sensors and gyroscope on an Arduino glove capture hand gestures; a self-collected, cleaned dataset was used to train a KNN classifier in Python, translating American Sign Language gestures into audible speech in real time. Demonstrates cross-disciplinary skills in ML, IoT, and embedded systems.

Tech: *Python, Arduino, Scikit-learn, K-Nearest Neighbour, IoT sensors*

Animal Image Classification System — [GitHub](#)

Designed a custom CNN architecture from scratch (no pre-trained weights) to classify 9 animal classes, achieving 79.51% accuracy. Key design choices: progressive convolutional depth, 3×3 filters, batch normalisation, and dropout regularisation to control overfitting.

Tech: *Python, PyTorch, OpenCV, NumPy, Scikit-learn*

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, MATLAB
ML / AI	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Open3D
Backend	Spring Boot, Django, REST APIs, Java Spark, NATS (event-driven messaging)
Frontend	Vue.js, HTML/CSS, JavaScript
Databases	MySQL, SQLite
Tools	Git, GitHub, Docker (foundational), Postman, Swagger, Linux, Virtual Environments
Concepts	Distributed Systems, Computer Vision, Deep Learning, Point Cloud Processing, API Design, Network Security

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Data Analytics Specialisation — Coursera (May 2023)
- API Designer & API Security Architect — API Academy
- AWS Academy Cloud Foundations — Amazon Web Services
- Deep Learning Onramp & Machine Learning Onramp — MathWorks
- Data Science Foundations — LinkedIn Learning